User Manual: Property IQ Deal Analyzer

This manual explains every metric, formula, and input in the RealEstate-Analytics.ai Property IQ Deal Analyzer, built for residential investment property: single-family rentals, 2–4 unit buildings, and small multifamily. It shows how to read a deal the way an investor and a lender will — return on cash invested, debt coverage at today's interest rates, multi-year cash flow, and the full return including the eventual sale — so you can walk a client through any property and defend every number.

1. Core Investment Metrics

Internal Rate of Return (IRR)

IRR is your annualized return over the whole hold — one rate that accounts for when each dollar comes back. Two versions are shown:

IRR Type	What It Measures
IRR (Operational)	Return from annual cash flows only — rent minus expenses minus mortgage. Excludes sale proceeds.
IRR (Total incl. Sale)	Adds the projected sale value — purchase price grown at your appreciation rate — to the final-year cash flow.

Tip: *Use Operational IRR to compare rental efficiency across deals. Use Total IRR to evaluate the full lifecycle return, including your exit.*

Cash-on-Cash Return (CoC)

Measures annual pre-tax cash flow relative to the initial cash invested.

Formula: CoC = (Annual Pre-Tax Cash Flow / Initial Cash Invested) × 100

Cap Rate

Capitalization rate measures a property's income yield independent of financing.

Formula: Cap Rate = (NOI / Purchase Price) × 100

NOI (Net Operating Income) = Effective Rent – Operating Expenses. Cap rate does not include mortgage payments — it is a property-level metric, not an equity metric.

Equity Multiple

Formula: Equity Multiple = Total Cash Inflows / Total Cash Invested

Measures how many times your original equity has grown over the hold period. Example: invest $100K, receive $220K total (rent + sale) = 2.2x equity multiple.

DSCR — Debt Service Coverage Ratio

Formula: DSCR = NOI / Annual Debt Service (mortgage payments)

DSCR	Signal
≥ 1.25×	Lender-approved — standard minimum for most commercial loans
1.0–1.25×	Borderline — some lenders accept, higher rates likely
< 1.0×	Below breakeven — NOI does not cover debt service

ROI (Return on Investment)

Formula: ROI = (Total Return / Initial Investment) × 100

ROI = (cumulative cash flow + appreciation) / down payment. It is cumulative and grows each year — this is the green ROI line on the Multi-Year Cash Flow chart.

2. Cash Flow & Multi-Year Projections

How Cash Flow Is Calculated

Formula: Cash Flow = Annual Effective Rent – Operating Expenses – Mortgage Payments

Effective Rent = Scheduled Rent × (1 – Vacancy Rate). This is the rent actually collected after vacancy and credit loss.

Vacancy Rate

The Vacancy Rate slider is treated as economic vacancy: it captures both empty months and rent that goes uncollected. Effective rent = scheduled rent × (1 – Vacancy Rate).

Tip: *If unsure, set Vacancy Rate to 5% for a stabilized rental, 8–10% for a value-add property.*

The Multi-Year Cash Flow Projection Chart

The chart plots three lines over your hold period:

Line	Color	What It Shows
Multi-Year Cash Flow ($)	Blue	Annual net cash flow after all expenses and debt service
Pro Forma Rent ($/yr)	Orange	Projected gross annual rent growing at your rent growth rate
ROI (%)	Green	Cumulative return on equity including appreciation, compounding each year

The left Y-axis shows dollar values (cash flow and rent); the right Y-axis shows ROI percentage, so all three trajectories are visible on one chart.

Investment Time Horizon (Years)

The number of years you plan to own the property before sale. Affects IRR (Total), Equity Multiple, ROI, and the terminal sale value used in all exit-based calculations.

3. Operating Expenses (OpEx)

Monthly operating expenses include all recurring costs the landlord is responsible for:

Expense Category	Notes
Property Taxes	Largest component in most markets. Verify with the county assessor.
Property Insurance	Hazard + liability. Typically $100–400/mo for residential, higher for commercial.
Maintenance & Repairs	Budget 0.5–1% of property value annually for residential.
HOA Fees	Where applicable. Fixed monthly charge.
Property Management	If third-party managed: typically 8–12% of gross rent.
Miscellaneous	Landscaping, pest control, professional fees, etc.

Tip: *If unsure of total OpEx, estimate 25–30% of gross monthly rent.*

4. IRR (Operational) vs. IRR (Total incl. Sale)

IRR Type	Cash Flows Used	Best For
IRR (Operational)	[–Down Payment, CF_1, CF_2, …, CF_n] — no sale proceeds	Comparing rental efficiency. Lender underwriting.
IRR (Total incl. Sale)	[–Down Payment, CF_1, CF_2, …, CF_n + Sale Value]	Full lifecycle return. Investor decision-making.

Illustrative Example (5-year hold)

Input / Result	Value
Down Payment	$40,000
Annual Net Cash Flow (Years 1–5)	$3,600
Net Exit Proceeds (Year 5)	$60,000
IRR (Operational)	≈ –21.8%
IRR (Total incl. Sale)	≈ 16.2%

Tip: *Over a short hold, cash flow alone never returns the $40,000 down payment, so Operational IRR is negative. The exit is what makes this deal work — which is exactly why both numbers are shown side by side.*